

CHAPTER - 8

Object Oriented Programming Language

> Object oriented programming tries to map code instructions with real world making the code short and easier to understand.

• What is object oriented programming?

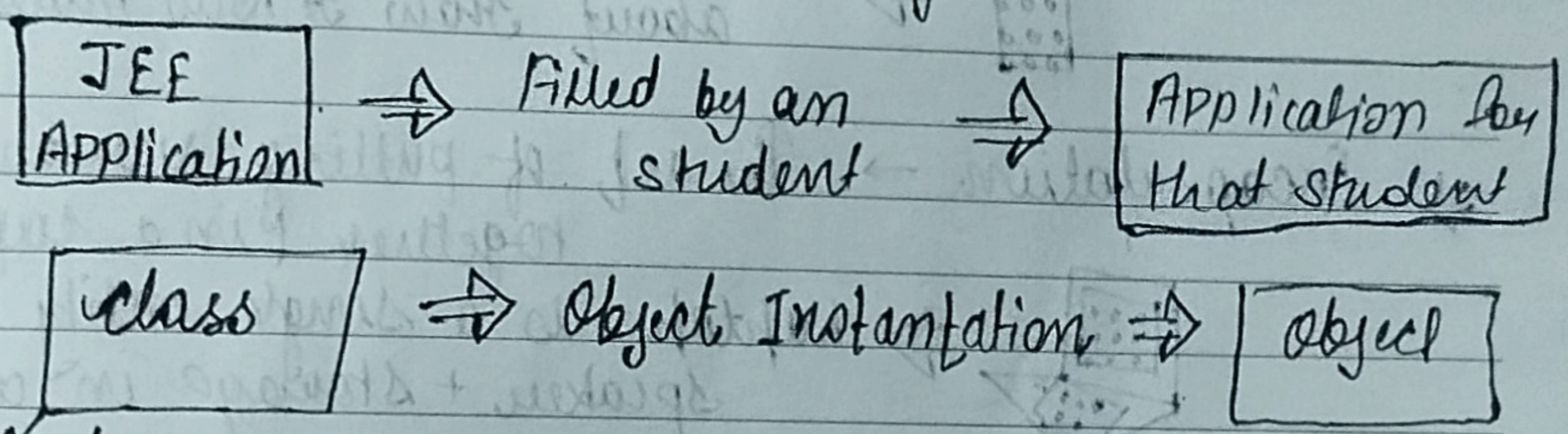
> Solving a problem by creating objects is one of the most popular approaches in programming. This is called object oriented programming.

• What is DRY?

> DRY stands for - Do not repeat yourself
↳ focuses on code reusability.

Class

A class is a blueprint for creating objects.



contains info to create a valid object

● Object

An object is an instantiation of a class when a class is defined, a template (info) is defined. Memory is allocated only after object instantiation.

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How to model a Problem in OOPs

Noun → class → Employee

Adjective → Attributes → name, age, salary.

Verb → Methods → getSalary(), increment().

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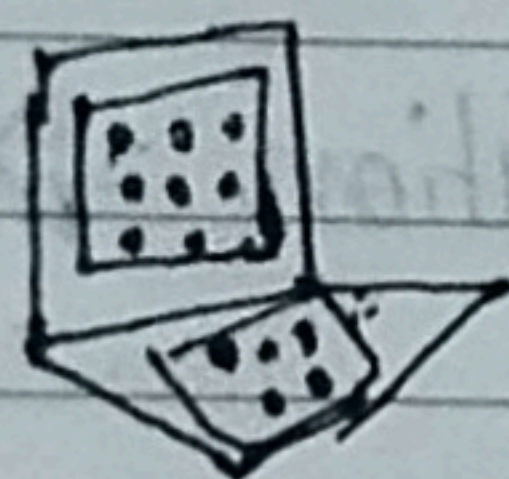
● OOPs Terminology

1) Abstraction → Hiding internal details (show only essential info!)



⇒ use this phone without bothering about how it was made.

2) Encapsulation → The act of putting various components together (in a capsule).



⇒ laptop is a single entity with wifi + speaker + storage in a single box!

~~3) Information~~ →

In Java, Encapsulation simply means that the sensitive data can be hidden from the users.

3) Inheritance → The act of deriving new things from existing things.

Rickshaw ⇒ E-Rickshaw
phone ⇒ Smart phone.

4) Polymorphism → One entity many forms.

Smartphone → Phone.

Smartphone → Calculator

● Writing a custom class

We can write a custom class as follows:

```
public class Employee
```

```
{
    int id;
```

→ Attribute 1

```
    String name;
```

→ Attribute 2

```
}
```

➤ A class with Methods

We can add methods to our class Employee as follows:

```
public class Employee {
```

```
    public int id;
```

```
    public String name;
```

```
    public int getSalary() {
```

```
        // code
```

```
    }
```

```
    public void getDetails() {
```

```
    }
```

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✓ Examples with code

```
public class Employee
```

```
{
```

```
    int id;
```

```
    String name;
```

```
    int salary;
```

```
    public void getDetails()
```

```
{
```

```
        System.out.println("The id of employee is: " + this.id);
```

```
        System.out.println("The name of employee is: " + this.name);
```

```
        System.out.println("The name salary of employee is: " + this.salary);
```

```
}
```

```
public static void main (String args)
```

```
{
```

```
    System.out.println("This is our custom class");
```

```
    Employee name1 = new Employee();
```

```
    Employee name2 = new Employee();
```

```
    name1.name = "Mahip";
```

```
    name1.id = 61;
```

```
    name1.salary = 70000;
```

```
    name2.name = "Mohit";
```

```
    name2.id = 59;
```

```
    name2.salary = 50000;
```

```
    name1.getDetails();
```

```
    name2.getDetails();
```

```
}
```